

The Utility Problem

Capacity is important in a distribution grid because it determines the amount of electrical power that can be supplied to customers. It is determined by multiple attributes, for example, the size and rating of the conductors and transformers used in the feeder line, as well as by the loading conditions, including the number of customers connected to the feeder, the load density, and the expected maximum demand.

If the capacity of the distribution grid is not sufficient, it can result in the overloading of the system, leading to power quality issues and even power outages. This can impact the reliability and quality of service of the distribution grid for customers and may also result in equipment damage.

Therefore, it is important to have adequate capacity in a distribution grid to ensure that there is enough electrical power available to meet the current and future demand for electricity. This can be done by upgrading existing infrastructure, adding new feeders or laterals, or by optimizing the use of existing capacity through demand management and load balancing.

The activities involving grid upgrades, or adding new infrastructure are always accompanied by higher costs and thus are more expensive. The less expensive option is to analyze feeders and feeder's laterals to find out where the loads exist and where they pose the biggest bottlenecks. For these reasons, it is important to have a system in place which allow for such kind of analysis.

The Data Solutions and Methods

In the North American grid at the medium-voltage level, there is typically a three-phase feeder from the transformer, and then single-phase feeders (laterals) are routed directly to customers via taps from this feeder. To obtain information about the capacity in certain grid areas, the capacity must be calculated for both feeder and laterals.

Data required for the calculation:

1. GIS data and tracing - GIS data allows to identify which sections belong to each feeder and which consumption points belong to the feeder and these sections.
2. Asset data - the maximum possible load capacity (from cross-section and conductor type) must be assigned to each feeder section. Each section may have a different cross-section and conductor type - typically, at the transformer, there will be a higher cross-section as it is necessary to supply power to all the consumption points at the feeder, and further away from the transformer, the cross-section may decrease as there is no longer a need to supply power to as many consumption points as at the beginning of the feeder - this is known as the economic cross-section.

3. Load on the feeder and its individual sections - the load on the feeder can be determined using time series data of the individual consumption points. These consumption points are summed for each section according to the topology.

Finally, the load of each section is compared along with the capacity of its conductor. This allows us to know the maximum load of each section and thus to know whether the section needs to be reinforced or whether it still has sufficient spare capacity.

Awesense Energy Transition Platform is ready for such analysis. By integrating a wide variety of data, all the necessary data is available in one place (the data is cleaned and synchronized) and is easy to access thanks to sophisticated functions for, for example, grid tracing and exports based on this tracing or time series data aggregation. With the use of the Energy Data Model's (EDM) APIs, it is possible to call the required endpoints and easily obtain the necessary information for such analysis.

A high level approach to such analysis would be:

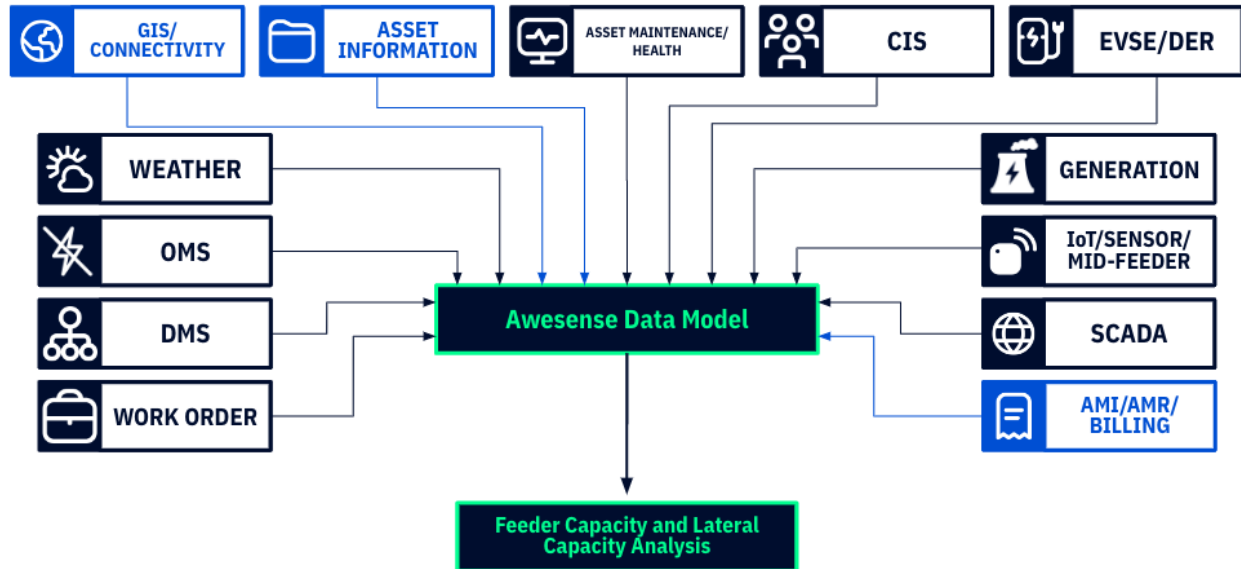
1. Choose the grid area which should be analyzed.
2. Trace the feeders using the `grid_get_downstream()` function to obtain the list of loads on each feeder.
3. Merge the information with loads phase information.
4. Get the load information using the `ts_data_source()` select function.
5. Obtain the capacity information from the asset data.
6. Compare the capacity information with loading information.
7. Evaluate the results and make recommendations.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Feeder Capacity and Lateral Capacity Analysis

[UC46-01] - Awesense Energy Transition Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.